

UNIVERSIDAD DEL EL SALVADOR
FACULTAD DE INGENIERÍA Y ARQUITECTURA
ESCUELA DE INGENIERÍA DE SISTEMAS INFORMÁTICOS
COMUNICACIONES 1 - COS115



CICLO 02 – 2026



TEMA:

SOLUCIÓN LABORATORIO PRACTICO 2

PRESENTADO POR:

VÁSQUEZ MENJÍVAR, LUIS ENRIQUE - VM15037

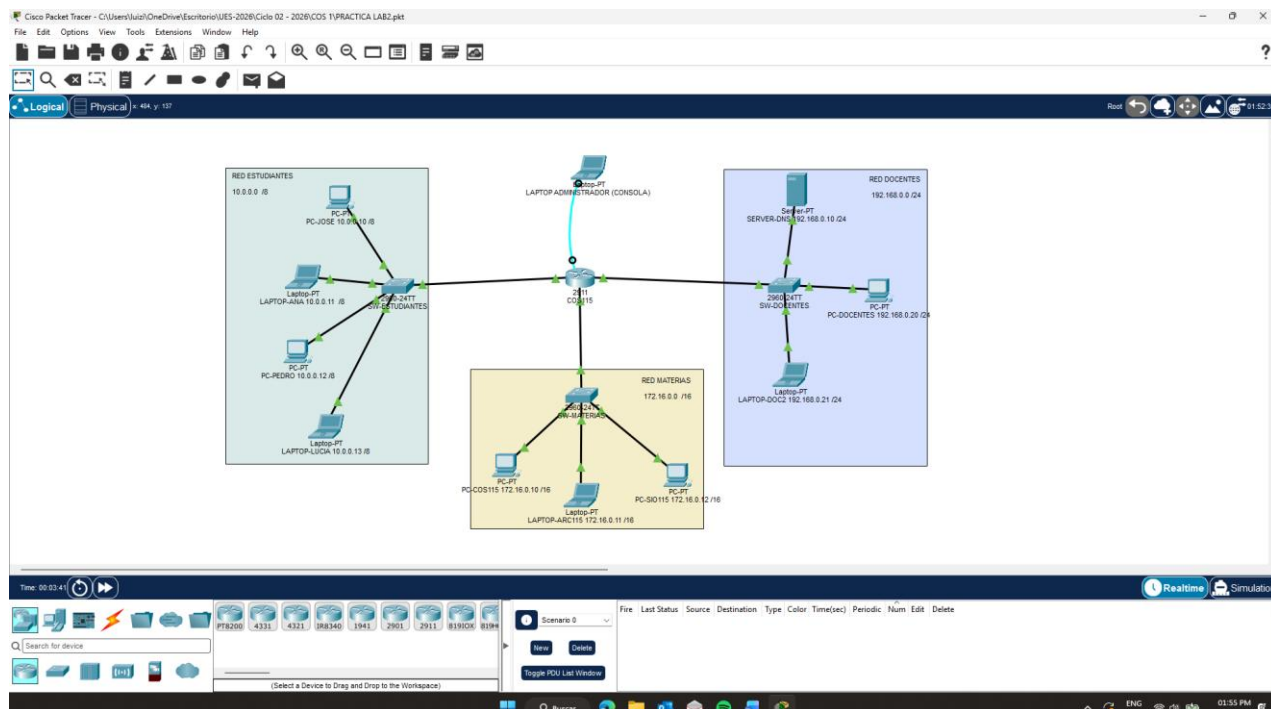
TUTOR RESPONSABLE:

ING. LIGIA MARCELA LAZO MAJANO

CIUDAD UNIVERSITARIA, SEPTIEMBRE DEL 2026

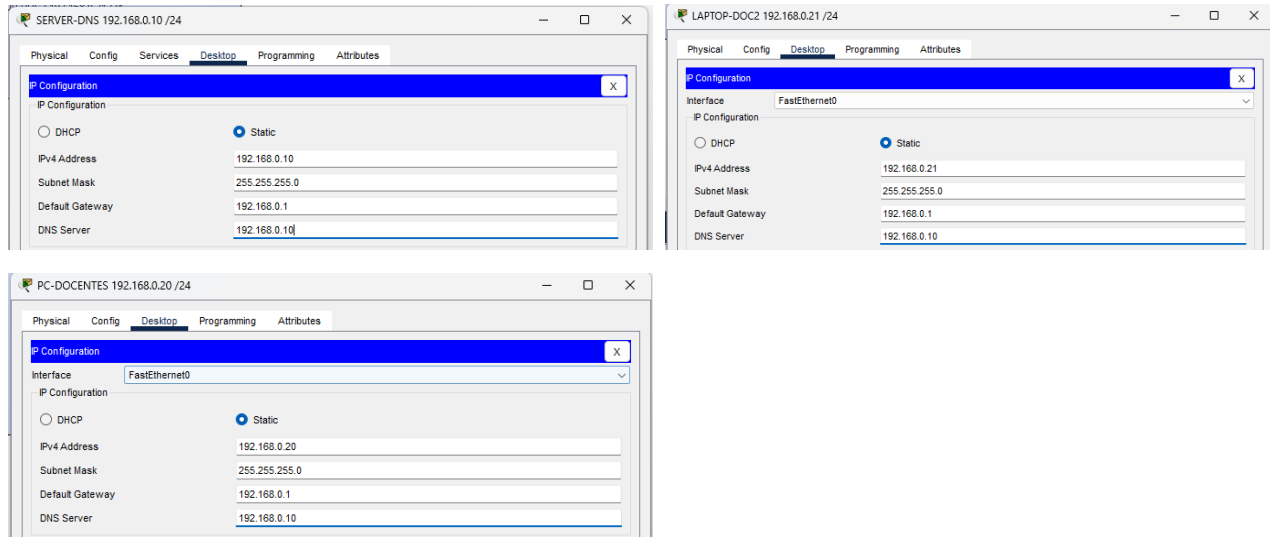
Captura de pantalla de la topología completa y correctamente conectada

A continuación, se deja constancia de las configuraciones y topología del sistema



Configuración de red DOCENTES en router y equipos junto con el Listado de los principales comandos utilizados:

```
Router>
Router>enable
Router#con
Router#conf
Router#configure t
Router#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#host
Router(config)#hostname COS115
COS115(config)#h
COS115(config)#hostname COS2026
COS2026(config)#
COS2026(config)#interface gig
COS2026(config)#interface gigabitEthernet 0/2
COS2026(config-if)#ip a
COS2026(config-if)#ip ad
COS2026(config-if)#ip address 192.168.0.1 255.255.255.0
COS2026(config-if)#des
COS2026(config-if)#description la
COS2026(config-if)#description lan
COS2026(config-if)#description lan DOCENTES
COS2026(config-if)#no s
COS2026(config-if)#no sh
COS2026(config-if)#no shutdown
COS2026(config-if)#
%LINK-5-CHANGED: Interface GigabitEthernet0/2, changed state to up
%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/2, changed state to up
COS2026(config-if)#
COS2026#
%SYS-5-CONFIG_I: Configured from console by console
```



Configuración de red MATERIAS en router y equipos junto con el Listado de los principales comandos utilizados:

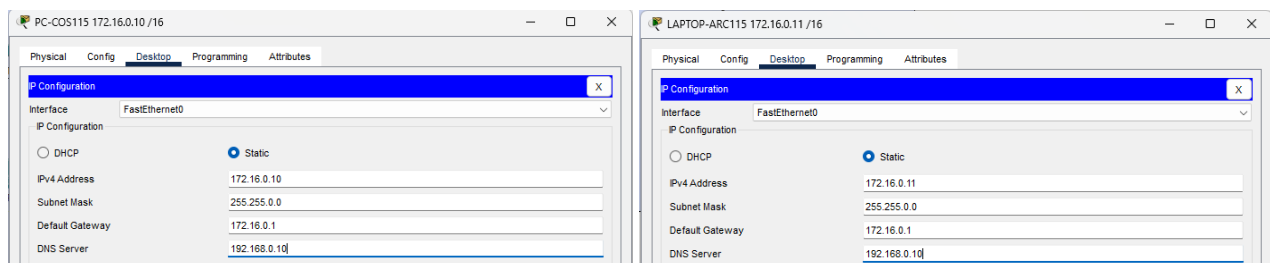
```

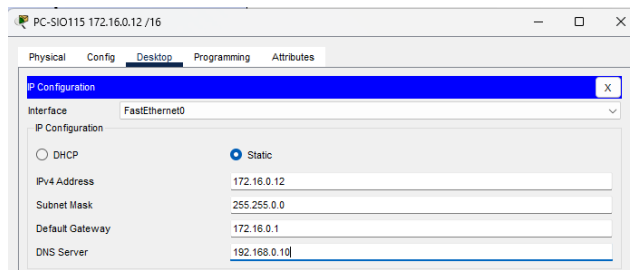
COS2026#con
COS2026#conf
COS2026#configure t
COS2026#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
COS2026(config)#in
COS2026(config)#interface gig
COS2026(config)#interface gigabitEthernet 0/0
COS2026(config)#interface gigabitEthernet 0/0
^
Invalid input detected at '^' marker.

COS2026(config)#interface gigabitEthernet 0/0
COS2026(config-if)#ip add
COS2026(config-if)#ip address 172.16.0.1 255.255.0.1
Bad mask 0xFFFF0001 for address 172.16.0.1
COS2026(config-if)#ip address 172.16.0.1 255.255.0.0
COS2026(config-if)#de
COS2026(config-if)#des
COS2026(config-if)#description 1s
COS2026(config-if)#description 1sn MATERIAS
COS2026(config-if)#no sh
COS2026(config-if)#no shutdown

COS2026(config-if)#
*LINK-5-CHANGED: Interface GigabitEthernet0/0, changed state to up
*LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/0, changed state to up

```





Configuración de red ESTUDIANTES en router y equipos junto con el Listado de los principales comandos utilizados:

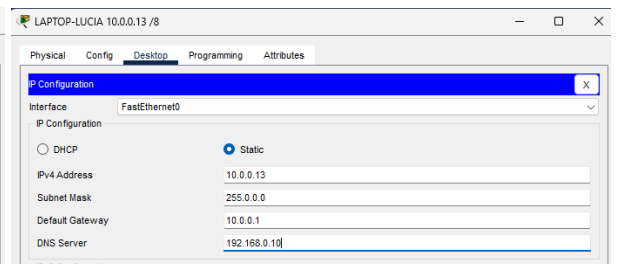
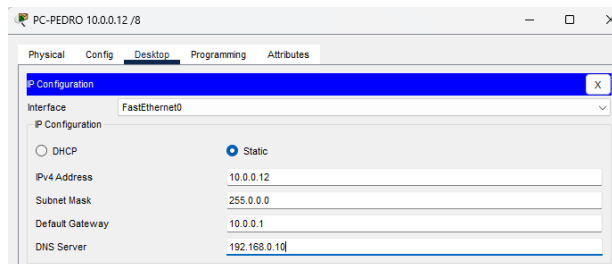
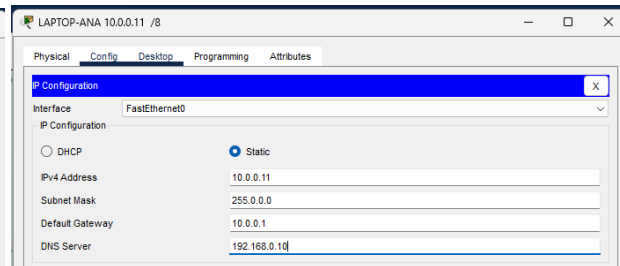
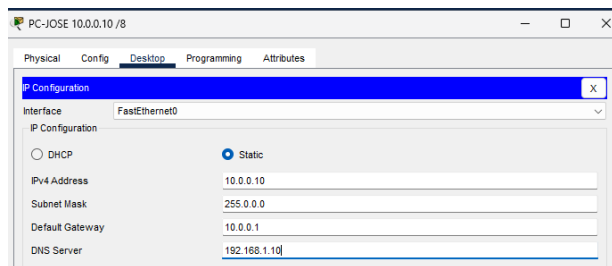
```

COS2026(config-if)#exit
COS2026(config)#gi
COS2026(config)#in
COS2026(config)#interface gi
COS2026(config)#interface gigabitEthernet 0/1
COS2026(config-if)#ip ad
COS2026(config-if)#ip address 10.0.0.1 255.0.0.0
COS2026(config-if)#des
COS2026(config-if)#description lan ESTUDIANTES
COS2026(config-if)#no
COS2026(config-if)#no s
COS2026(config-if)#no sh
COS2026(config-if)#no shutdown

COS2026(config-if)#
%LINK-5-CHANGED: Interface GigabitEthernet0/1, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/1, changed state to up

```



RESUMEN DE CONFIGURACIONES DE IP ROUTER

```
IOS Command Line Interface
563859 packets output, 0 bytes, 0 underruns
0 output errors, 23 interface resets
0 output buffer failures, 0 output buffers swapped out

COS2026#
COS2026#
COS2026#
COS2026#
COS2026#
COS2026#
COS2026#
COS2026#
COS2026#
COS2026#sh
COS2026#show ip
COS2026#show ip r
COS2026#show ip ro
COS2026#show ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
       i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
       * - candidate default, U - per-user static route, o - ODR
       P - periodic downloaded static route

Gateway of last resort is not set

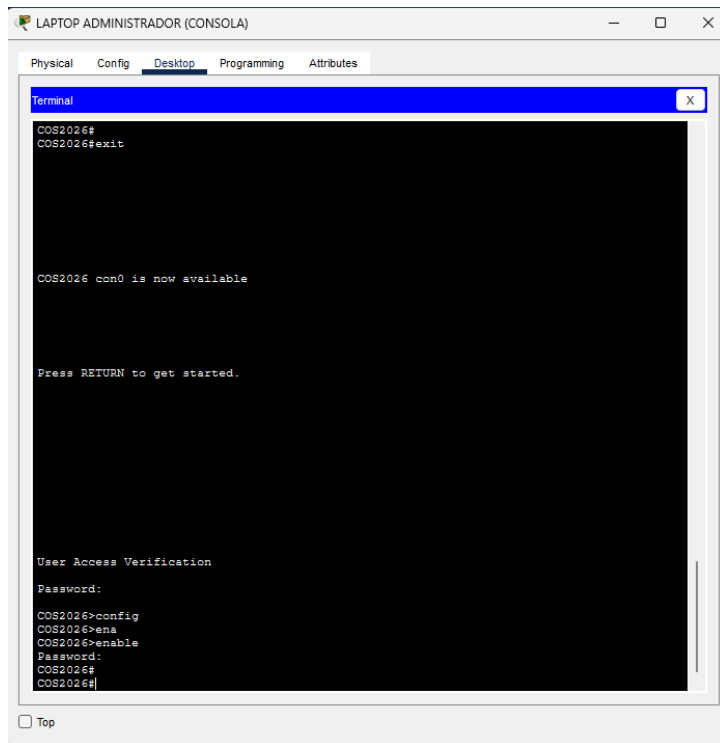
10.0.0.0/8 is variably subnetted, 2 subnets, 2 masks
C    10.0.0.0/8 is directly connected, GigabitEthernet0/1
L    10.0.0.1/32 is directly connected, GigabitEthernet0/1
172.16.0.0/16 is variably subnetted, 2 subnets, 2 masks
C    172.16.0.0/16 is directly connected, GigabitEthernet0/0
L    172.16.0.1/32 is directly connected, GigabitEthernet0/0
192.168.0.0/24 is variably subnetted, 2 subnets, 2 masks
C    192.168.0.0/24 is directly connected, GigabitEthernet0/2
L    192.168.0.1/32 is directly connected, GigabitEthernet0/2

COS2026#
```

Capturas de las pruebas de conectividad y protocolo de acceso SSH.

```
COS2026#configure t
COS2026#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
COS2026(config)#ena
COS2026(config)#enable se
COS2026(config)#enable secret COS02
COS2026(config)#li
COS2026(config)#lin
COS2026(config)#line console
COS2026(config)#line console 0
COS2026(config-line)#pa
COS2026(config-line)#pas
COS2026(config-line)#password COS01
COS2026(config-line)#lo
COS2026(config-line)#log
COS2026(config-line)#logl
COS2026(config-line)#login
COS2026(config-line)#
COS2026#
*SYS-5-CONFIG_I: Configured from console by console
```

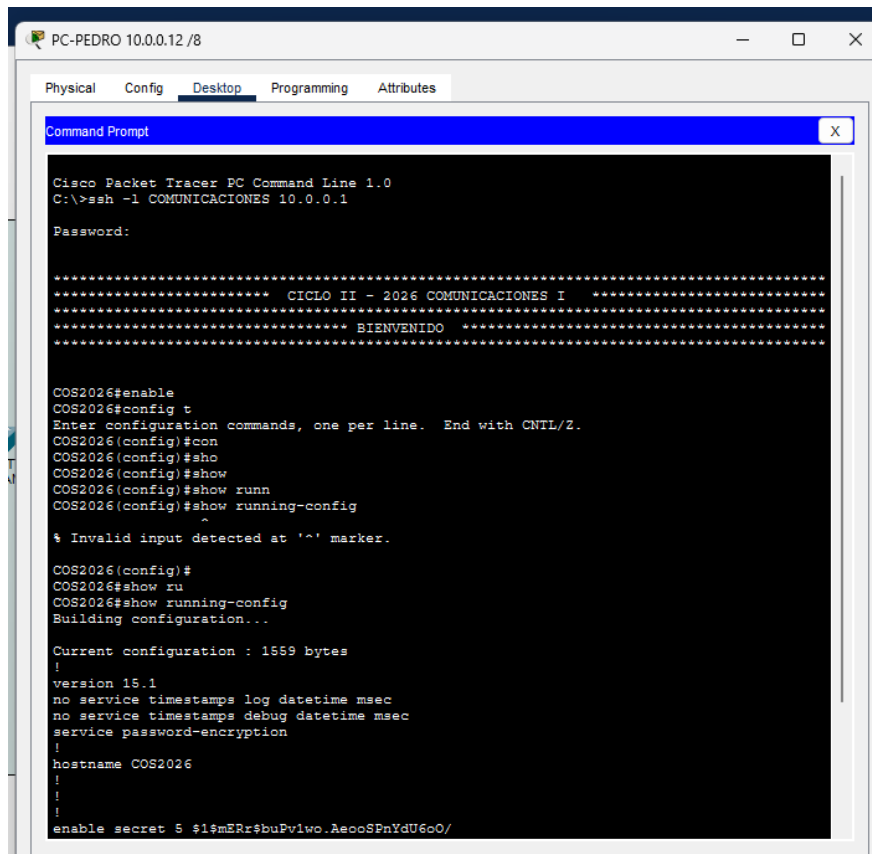
ACCESO POR CONSOLA DESPUES DE LAS CONFIGURACIONES



CONFIGURACIONES TELNET

```
COS2026(config)#
COS2026(config)#ip domain-na
COS2026(config)#ip domain-name comunicaciones2626.sv
COS2026(config)#crypto key generate rsa
COS2026(config)#crypto key generate rsa
The name for the keys will be: COS2026.comunicaciones2626.sv
Choose the size of the key modulus in the range of 360 to 4096 for your
General Purpose Keys. Choosing a key modulus greater than 512 may take
a few minutes.
How many bits in the modulus [512]: 1024
% Generating 1024 bit RSA keys, keys will be non-exportable...[OK]
COS2026(config)#ip ssh v
*Mar 1 3:0:8.67: %SSH-5-ENABLED: SSH 1.99 has been enabled
COS2026(config)#ip ssh version 2
COS2026(config)#line v
COS2026(config)#line vty 0 15
COS2026(config-line)#login lo
COS2026(config-line)#login local
COS2026(config-line)#tran
COS2026(config-line)#transport in
COS2026(config-line)#transport input ssh
COS2026(config-line)#user
COS2026(config-line)#userb
COS2026(config-line)#usern
COS2026(config-line)#usernam
COS2026(config-line)#username COMUNICACIONES pri
COS2026(config-line)#username COMUNICACIONES priv
COS2026(config-line)#username COMUNICACIONES privi
COS2026(config-line)#username COMUNICACIONES privilege 15 secret COS115
COS2026(config)#servi
COS2026(config)#service pas
COS2026(config)#service password-encryption
COS2026(config)#
```

ACCESO TELNET DESPUES DE LAS CONFIGURACIONES



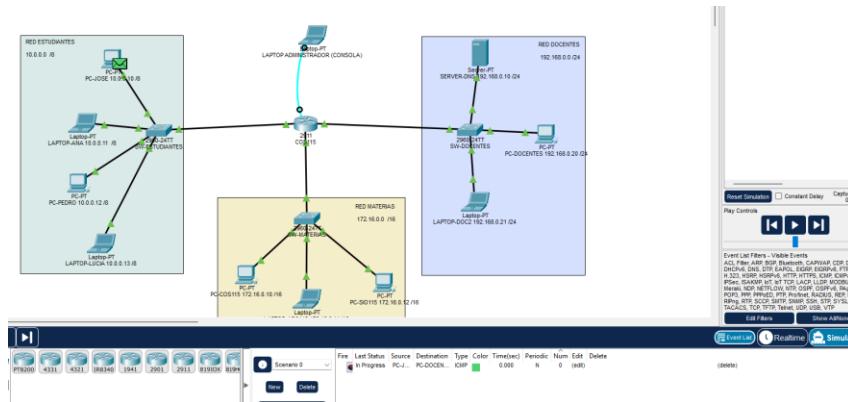
CREACION DEL BANNER EN ROUTER

```
COS2026(config)#ba
COS2026(config)#banner motd #
Enter TEXT message. End with the character '#'.
*****
*****  CICLO II - 2026 COMUNICACIONES I  *****
*****
*****  BIENVENIDO  *****
*****
#
COS2026(config)#
```

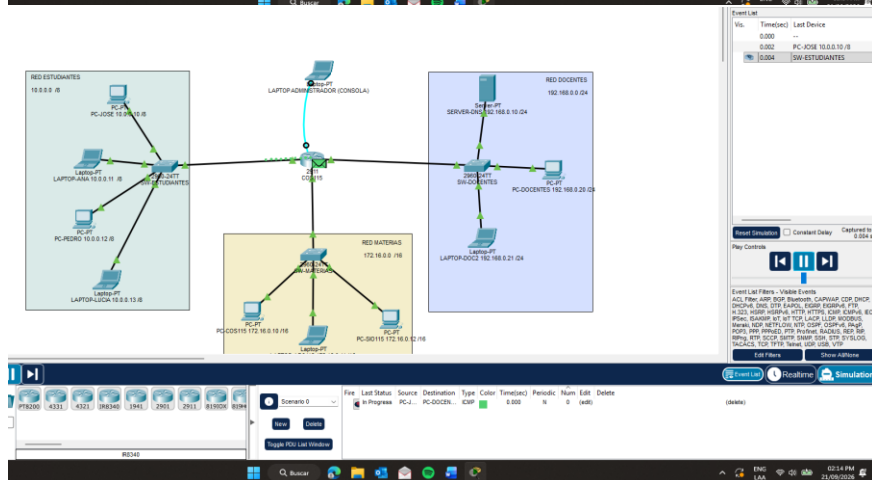
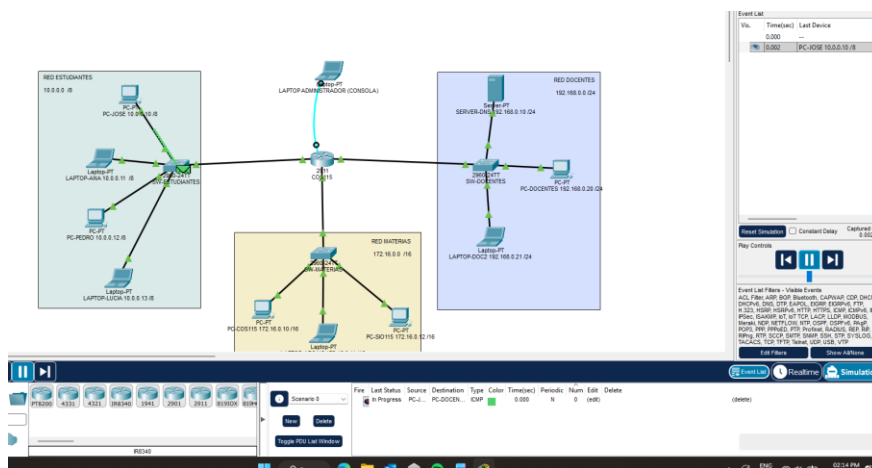
PRUEBAS REALIZADAS Y SUS RESULTADOS.

Se realiza la simulación de envío de un paquete entre redes de manera grafica para verificar como el Router comunica a la red Estudiantes (10.0.0.0 /8) con la red Docentes (192.168.0.0 / 24)

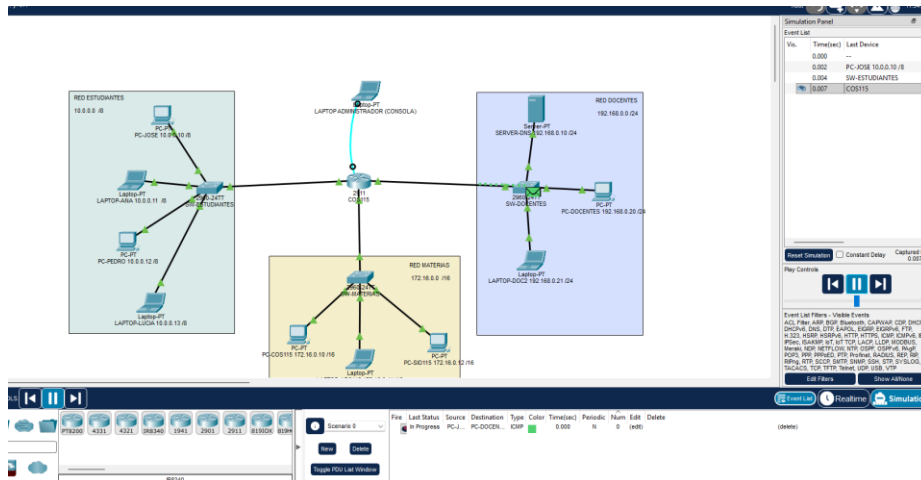
1. Se comienza seleccionando un paquete ICMP desde la PC de estudiante Jose Hasta la PC de Docente



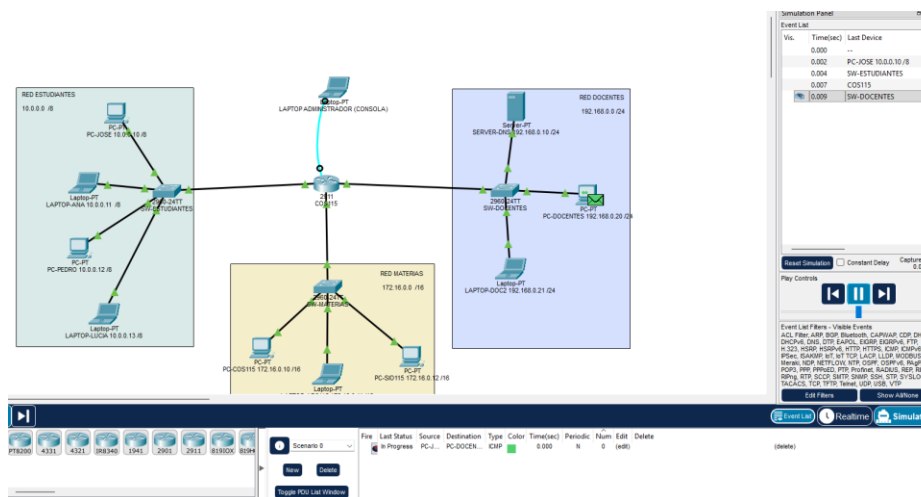
2. Al detectar que la IP de destino (192.168.0.x) no pertenece a su red (10.0.0.0/8), la PC envía la trama directamente a la dirección MAC de su Puerta de Enlace (el Router). El Switch simplemente recibe la trama, lee que va dirigida a la MAC del Router y la reenvía por el puerto correspondiente.



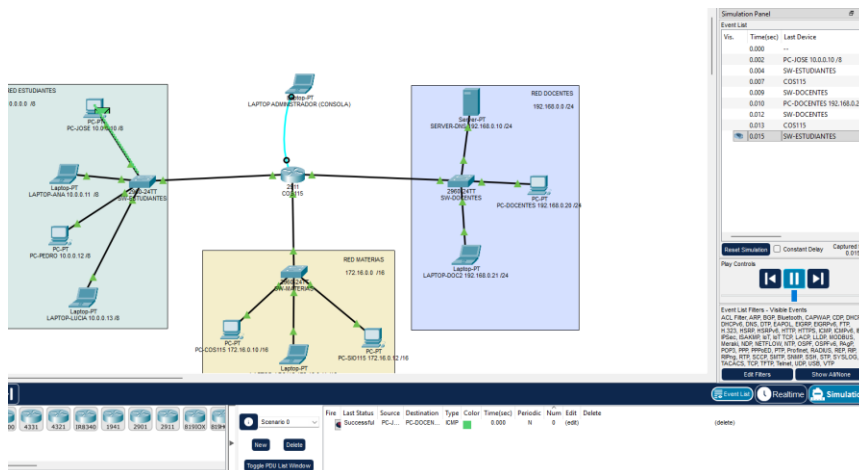
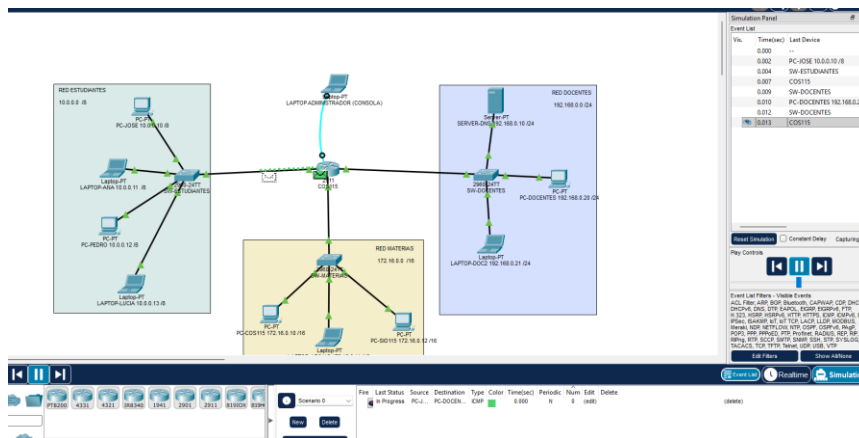
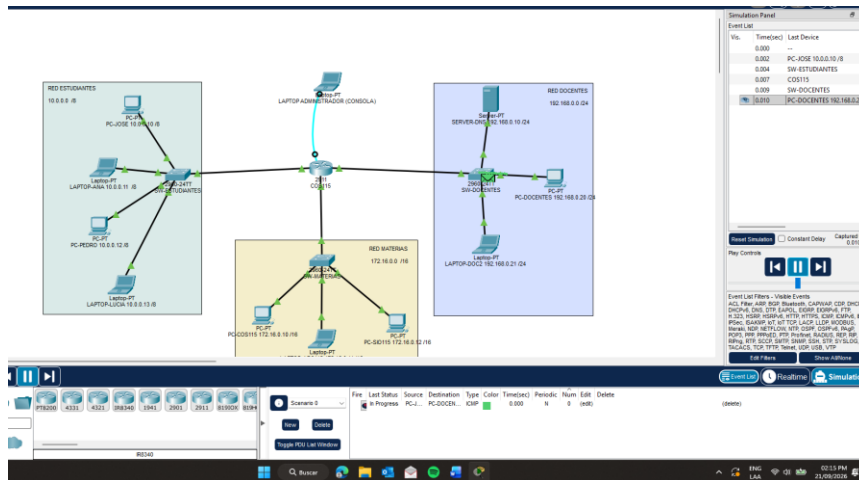
- El router desencapsula el paquete, revisa la tabla y lo vuelve a encapsular para la nueva red. revisa sus tablas de capa 3 de ruteo y envía el paquete al puerto donde se registro la IP del destino



- El Switch recibe el paquete en capa 2 y lo envía revisando el destino por medio de su tabla de CAM (registro de MAC conectadas) registrada



5. El destino recibe el paquete y envia una respuesta indicando que el paquete fue recibido exitosamente, el cual sigue la misma ruta antes mencionada. *ICMP Echo Reply*



Prueba desde el CMD

Se realiza un ping desde la maquina con IP 10.0.0.11 hacia las 3 redes conectadas al router

The image displays a network simulation environment with a central router (R1) connected to three distinct networks. The top window shows a Command Prompt on a laptop (LAPTOP-ANA 10.0.0.11) performing ping tests to three destinations: 192.168.0.10, 172.16.0.10, and 10.0.0.13. The results show successful pings to 192.168.0.10 and 10.0.0.13, but a request timed out for 172.16.0.10. The bottom window shows the network topology, including a central router (R1) connected to three networks: RED ESTUDIANTES (10.0.0.0/8), RED MATEMATICAS (172.16.0.0/8), and RED FISICA (10.0.0.0/8). The router is also connected to a LAPTOP ADMINISTRADOR (CONSOLA). The bottom window also shows a list of devices and their IP addresses, including PC-JOSE, PC-PEDRO, PC-COS115, and PC-SD1151.

```
Command Prompt

C:\>ping 192.168.0.10

Pinging 192.168.0.10 with 32 bytes of data:

Reply from 192.168.0.10: bytes=32 time<1ms TTL=127
Reply from 192.168.0.10: bytes=32 time<1ms TTL=127
Reply from 192.168.0.10: bytes=32 time<1ms TTL=127
Reply from 192.168.0.10: bytes=32 time<1ms TTL=127

Ping statistics for 192.168.0.10:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 0ms, Average = 0ms

C:\>ping 172.16.0.10

Pinging 172.16.0.10 with 32 bytes of data:

Request timed out.
Reply from 172.16.0.10: bytes=32 time<1ms TTL=127
Reply from 172.16.0.10: bytes=32 time<1ms TTL=127
Reply from 172.16.0.10: bytes=32 time<1ms TTL=127

Ping statistics for 172.16.0.10:
    Packets: Sent = 4, Received = 3, Lost = 1 (25% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 0ms, Average = 0ms

C:\>ping 10.0.0.13

Pinging 10.0.0.13 with 32 bytes of data:

Reply from 10.0.0.13: bytes=32 time<1ms TTL=128
Reply from 10.0.0.13: bytes=32 time<1ms TTL=128
Reply from 10.0.0.13: bytes=32 time<1ms TTL=128
Reply from 10.0.0.13: bytes=32 time<1ms TTL=128

Ping statistics for 10.0.0.13:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 0ms, Average = 0ms

C:\>
```

RED ESTUDIANTES
10.0.0.0/8

PC-JOSE 10.0.0.10/8

Laptop-PT
LAPTOP-ANA 10.0.0.11/8

PC-PEDRO 10.0.0.12/8

PC-COS115 10.0.0.13/8

RED MATEMATICAS
172.16.0.0/8

PC-COS115 172.16.0.10/8

PC-SD1151

RED FISICA
10.0.0.0/8

PC-COS115 10.0.0.13/8

PC-SD1151

LAPTOP ADMINISTRADOR (CONSOLA)

Time: 00:34:06

Scenario: 0

Fire: Last Status

Success

Toggle PCUI List Window

Realtime Simulation